

ROCHA, Uruguay

WMO: 865650

Lat: 34.4935S

Lon: 54.3126W

Elev: 19

StdP: 101.10

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 1.0	(c) 2.3	(d) -1.1	(e) 3.5	(f) 4.2	(g) 0.1	(h) 3.8	(i) 4.7	(j) 10.4	(k) 13.6	(l) 8.9	(m) 12.9	(n) 0.4	(o) 350	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 10.6	(c) 31.5	(d) 22.4	(e) 30.0	(f) 22.0	(g) 28.6	(h) 21.6	(i) 24.2	(j) 28.9	(k) 23.4	(l) 27.7	(m) 22.7	(n) 26.6	(o) 3.6	(p) 0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	17.6	26.4	22.1	16.9	25.5	21.5	16.2	24.8	73.4	29.0	70.1	27.7	67.4	26.7	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	7.9	6.9	-1.4	35.6	0.8	1.9	-2.0	37.0	-2.4	38.1	-2.9	39.2	-3.4	40.6
			-1.5	25.7	0.8	1.2	-2.1	26.6	-2.6	27.3	-3.0	28.0	-3.6	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	22.2	21.6	19.8	17.4	14.1	11.6	10.9	12.1	12.9	15.5	17.8	20.5
	DBStd	5.08	2.91	2.83	2.90	3.18	3.20	3.40	3.73	3.72	3.36	3.10	3.14	3.36
	HDD10.0	83	0	0	0	0	3	19	33	19	9	1	0	0
	HDD18.3	1181	3	4	18	54	136	206	233	198	167	99	46	17
	CDD10.0	2397	377	326	305	222	131	66	61	83	96	170	236	326
	CDD18.3	452	122	97	64	26	5	2	3	4	4	10	31	84
	CDH23.3	3118	948	638	362	98	17	6	11	27	30	68	233	681
	CDH26.7	838	281	175	69	14	1	0	1	5	7	13	52	219
Wind	WSAvg	2.5	2.6	2.4	2.3	2.2	2.0	2.3	2.3	2.6	2.9	2.8	2.9	2.9
Precipitation	PrecAvg	1106	100	101	109	101	96	102	110	95	96	98	88	72
	PrecMax	1609	283	284	446	487	296	327	241	363	219	259	232	270
	PrecMin	656	6	4	12	11	19	3	16	17	25	18	26	3
	PrecStd	258	63	68	85	88	56	68	58	58	51	58	53	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.3	30.7	28.9	25.4	23.9	25.0	26.7	26.8	28.4	30.7	33.9
		MCWB	23.2	23.1	22.1	22.6	19.2	19.3	18.7	19.7	20.4	20.0	21.4	22.8
	2%	DB	31.3	30.6	28.5	26.0	22.8	21.0	20.9	22.8	22.7	25.0	28.1	31.0
		MCWB	22.7	22.8	21.4	20.7	17.9	17.6	17.0	17.4	17.1	18.8	20.5	21.4
	5%	DB	29.5	28.7	26.9	24.1	20.9	18.8	18.3	20.0	20.5	22.9	26.0	29.1
		MCWB	22.3	22.2	21.2	19.7	17.3	15.9	15.4	15.9	15.9	17.9	19.4	21.2
	10%	DB	28.0	26.9	25.3	22.6	19.4	17.1	16.5	17.9	18.6	21.0	24.0	27.1
		MCWB	21.8	21.4	20.5	19.2	16.7	14.7	14.3	15.0	14.9	16.8	18.6	20.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	25.2	24.3	23.1	20.3	20.2	19.2	20.5	21.3	21.5	23.1	24.7
		MCDB	30.0	30.3	27.9	27.3	23.6	23.1	23.5	25.1	26.2	26.1	28.4	30.7
	2%	WB	24.3	23.9	23.1	21.9	19.1	18.3	17.5	18.5	18.7	20.0	21.5	23.3
		MCDB	28.9	28.5	26.4	24.9	21.7	20.4	20.2	21.7	21.8	23.6	26.1	28.4
	5%	WB	23.5	23.1	22.2	20.9	18.2	16.8	16.2	16.9	17.0	18.7	20.4	22.3
		MCDB	27.6	27.0	25.2	22.9	20.1	18.5	17.9	19.0	19.0	21.4	24.2	26.8
	10%	WB	22.7	22.3	21.4	20.0	17.2	15.3	14.9	15.5	15.9	17.7	19.4	21.4
		MCDB	26.3	25.7	24.3	21.7	18.9	16.6	16.3	17.3	18.0	20.2	22.8	25.4
Mean Daily Temperature Range	5% DB	MDBR	10.6	9.7	9.8	9.0	9.2	9.1	8.9	9.0	9.5	9.6	10.5	11.0
		MCDBR	12.7	12.1	12.1	11.1	10.9	10.6	10.2	12.1	12.5	12.6	13.7	14.4
		MCWBR	5.7	5.7	6.0	6.2	6.6	7.1	6.5	7.1	7.4	7.2	7.2	6.7
	5% WB	MCDBR	9.8	9.9	9.4	7.9	8.8	8.2	8.1	9.8	9.2	9.8	10.9	11.4
		MCWBR	5.1	5.3	5.4	4.9	5.7	6.2	5.8	6.6	6.4	6.4	6.5	5.9
Clear-Sky Solar Irradiance	taub	0.399	0.403	0.373	0.373	0.360	0.363	0.370	0.449	0.463	0.416	0.387	0.387	
	taud	2.386	2.394	2.466	2.412	2.425	2.399	2.341	2.071	2.053	2.251	2.366	2.396	
	Ebn at Noon	938	912	903	840	793	757	769	742	797	889	943	954	
	Edh at Noon	128	122	106	99	87	83	93	137	157	139	129	127	
All-Sky Solar Radiation	RadAvg	7.13	6.09	5.04	3.72	2.62	2.22	2.31	3.03	4.15	5.39	6.73	7.26	
	RadStd	0.54	0.58	0.44	0.41	0.18	0.20	0.21	0.30	0.31	0.48	0.56	0.49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.48	+0.70	N/A	+0.69	+0.63	N/A	N/A	N/A	+125	+67

Nomenclature: See separate page